

SUNDAY IKECHUKWU

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RESEARCH INTEREST

Adaptive Control & Learning-Based Strategies, Autonomous Systems in Unstructured Environments, Sim-to-Real Transfer, Planning & Control under Uncertainty, Robust Autonomy.

EDUCATION

University of Benin, Benin City, Edo State, Nigeria 01/2020 - 01/2025
Bachelor of Engineering (B.Eng) in Mechatronics Engineering
Honors: First Class Honors (Distinction), Class Rank: 1 out of 120
CGPA: 4.61/5.0 (WES iGPA: 3.77/4.0 | Scholaro: 3.796/4.0)

RESEARCH EXPERIENCE

Graduate Research Assistant, University of Benin 05/2025 – Present
Mechanical Engineering Supervisor: Prof. Ariavie G.O.

- Project 1: “Towards Automated Maritime Logistics: Simulation-Based Design of a Robotic Arm for Container Ship Cargo Handling” (Mechatronics Lab)**
 - Led the mechatronics development for a 6-DOF robotic arm system for autonomously handling of containerized cargo ship.
 - Designed high-fidelity simulations in CoppeliaSim, implementing a custom inverse kinematics solver for heterogeneous cargo handling.
 - Mentoring two undergraduate researchers on simulation best practice and kinematics control strategies.
- Project 2: "A Vision-Based Approach to Mitigating Vandalism and Enhancing Leak Detection in Nigeria's Oil and Gas Industry"**
 - Developing a real-time anomaly detection pipeline using YOLO and thermal imaging for deployment on edge devices (UAVs)
 - Conducted a gap analysis of 50+ publications to optimize the architecture for low-latency inference in field operations.
 - Leading custom dataset creation and validation protocols, supporting model training, real-world testing, and preparation of research findings for publication.

Graduate Research Assistant – Computer Vision for Critical Infrastructure Monitoring 07/2025 - Present
University of Benin, Department of Computer Engineering Principal Investigator: Dr. Isi Edeoghon

Research Focus: Deep Learning frameworks for autonomous fault detection in electrical infrastructure, addressing critical challenges in preventive maintenance and grid reliability

- Designed and implemented a two-stage deep learning pipeline combining Mask R-CNN (instance segmentation) with hierarchical classification for precise fault localization in complex aerial imagery.
- Solved the challenge of class confusion in cluttered backgrounds by implementing targeted data augmentation protocols, directly improving mean Average Precision (mAP) scores
- Engineered the system to be robust against real-world complexities, specifically occlusions and illumination variations in aerial imagery.

Undergraduate Research Assistant, University of Benin 07/2024 – 02/2025
Mechatronics Lab Supervisor: Prof. Obanor A.I.
Co-Supervisor: Dr. Sylvester Akinbohun

- Designed and developed a proof-of-concept electronic voting system with fingerprint-based biometric authentication for secure voter identification.
- Validated the system through simulation, integrating custom hardware (Arduino/microcontrollers) with software for real-time data processing.

PUBLICATIONS & MANUSCRIPTS

Peer-Reviewed Journal Articles

- Ojefia, F. E., Arigbanla, A. A., **Sunday, I.**, Adetunji, E. O., Enodiana, O. M., & Ekwonu, F. T. (2025). Design of a Mini-Autonomous Vehicle for Smart Mobility Solutions. *World Journal of Advanced Research and Reviews*, 28(1), 434–448. <https://doi.org/10.30574/wjarr.2025.28.1.3443>

Manuscripts in Preparation

- **Ariavie, G.O., Sunday, I.,** Obinna, O., Ogbesobor, I., & Martins, S. *Towards Automated Maritime Logistics: Simulation-Based Design of a Robotic Arm for Container Ship Cargo Handling.* (Prepared for submission).
- **Ariavie, G.O.,** Simon, T. A., & **Sunday, I.** *A Vision-Based Approach to Mitigating Vandalism and Enhancing Leak Detection in Nigeria's Oil and Gas Industry.* (Manuscript in preparation).

TEACHING/ACADEMIC EXPERIENCE

Teaching Assistant, Department of Mechatronics Engineering 07/2025 – Present
University of Benin, Benin City, Edo State, Nigeria

- **Technical Evaluation:** Assessing and grading technical logbooks and industrial reports for final-year (500L) students returning from the Student Industrial Work Experience Scheme (SIWES), ensuring alignment with engineering standards
- **Examination Assessment:** Assisted Prof. G.O. Ariavie in grading examinations for core mechatronics course, ensuring accurate and consistent assessment of student performance for 100+ undergraduate students.
- Facilitated laboratory sessions for undergraduate students, providing guidance on robotics simulation tools and control system fundamentals.

Technical Instructor, Chestnut and Bolt Technology, Benin City Nigeria 11/2024 – 01/2025

- Equipped interns with foundational skills in electronics, Embedded C, and Arduino IDE through hands-on training. Enabling them to transition from theoretical knowledge to practical prototyping projects within weeks.

Undergraduate Tutor, University of Benin, Benin City, Edo State, Nigeria 01/2021 – 12/2024

- Provided peer-tutoring to over 70 undergraduate students in core courses including “Robotics and Automation”, Introduction to robotics, Control Engineering, helping ~80% of participating students secure “A” grades in these challenging courses.

PROFESSIONAL EXPERIENCE

Embedded System Intern, Chestnut & Bolt Technologies | Edo State, Nigeria 01/2024 - 05/2024

- Designed and deployed a prototype pick-and-place robotic arm system, integrating PIC/Arduino microcontrollers with custom hardware.
- **Critical Contribution:** Identified and documented a sim-to-real reliability gap in grasping performance caused by lighting and friction variance — an experience that directly shaped my research focus on robust, adaptive control policies for real-world deployment.
- Designed and simulated embedded control circuits in Proteus to validate system logic prior to physical prototyping.
- Collaborated with the engineering team to automate control systems, enhancing practical skills in system integration and component selection.

SKILLS

Technical Skills

Robotics & Simulation: ROS2 (Nav2, SLAM), MoveIt, CoppeliaSim, Gazebo, Linux(Ubuntu)

AI & Computer Vision: PyTorch, Roboflow, OpenCV, YOLO, Mask R-CNN, Reinforcement Learning, Google Colab

Programming: C++, Python, Embedded C (Intermediate), Arduino Programing

Hardware & Electronics: Arduino, Circuit Design & Simulation using Proteus, Soldering components onto PCBs

Soft Skills: Strong Research Writing Skills Leadership and Teamwork, Problem-Solving and Critical Thinking, Analytical Ability, Communication and Interpersonal Skills, Time Management, Adaptability and Continuous Learning

LEADERSHIP & VOLUNTEERING EXPERIENCE

SECRETARY GENERAL, Mechatronics Students Association, University of Benin 03/2024 – 10/2024

- Coordinated operations, managed records, and organized academic workshops, strengthening collaboration among engineering students.

MEMBER OF PARLIAMENT, National Association of Marine, Mechanical and Mechatronics Engineering Students (NAMES)- UNIBEN Chapter, University of Benin 03/2024 – 10/2024

- Represented mechatronics students in parliamentary sessions, contributing to policy discussions and decision-making on student welfare and academic matters.

Science & Math Club Mentor (Volunteer)- Chrisidia Education Center, Benin City.

05/2025 – 07/2025

- Guided the school’s Science and Math Club, introducing over 300 students to innovative STEM concepts beyond the classroom.
- Organized mathematics quizzes and collaborative problem-solving sessions that boosted student engagement and performance.
- Introduced robotics and Arduino programming through hands-on demonstrations, sparking early interest in engineering and technology among high school students.

Academic Mentorship, University of Benin

05/2023 – Present

- Mentored junior undergraduate students in the Mechatronics Engineering program, providing guidance on effective approaches to challenging core courses and study strategies.
- Made course materials and academic support consistently available, resulting in measurable performance improvements, with several mentees now on track to graduate with First Class honors.

AWARDS AND HONOURS

• Best Graduating Student, Department of Mechatronics Engineering, University of Benin	04/2025
• First-Ever First Class Graduate, Department of Mechatronics Engineering University of Benin	04/2025
• Best Undergraduate Final Year Research Thesis, Mechatronics Engineering	02/2025
• Best 300Level Student, Department of Mechatronics Engineering, University of Benin	11/2023
• Monday Onyeme Foundation Scholarship Award	09/ 2023
• Best Technical Report in MTE381(Technical Report Writing & Presentation).	05/2023
• TotalEnergies/JDA Merit Scholarship	05 /2022
• Asiruwa Scholarship Foundation for Outstanding Academic Performance	01/2022
• Best 100Level Student across M3(Marine, Mechanical and Mechatronics) Engineering Departments	09/2021

PROFESSIONAL DEVELOPMENT AND CERTIFICATIONS

• Hugging Face Deep Reinforcement Learning Course	(In progress, started 10/2025)
• Modern Robotics, Course 2: Robot Kinematics –(Coursera)	10/2025
• Modern Robotics, Course 1: Foundations of Robot Motion – (Coursera)	09/2025
• UC Berkeley CS 285: Deep Reinforcement Learning (Instructor: Sergey Levine) – (YouTube)	(In progress, started 10/2025)
• AI for Robotics (CS 7638) – Georgia Tech (OMSCS Course Videos): Path planning, localization, and Kalman filters	09/2025
• Fundamentals of Manuscript Preparation, Elsevier Research Academy	09/2025
• Data Analysis with Python – freeCodeCamp	03/2025
• Supervised Machine Learning: Regression and Classification – Coursera (Andrew Ng, Stanford).	04/2024

PROFESSIONAL MEMBERSHIPS

- Regular Member, African Robotics Network AfROB
- Nigeria Institute of Mechanical Engineers, NiMechE.
- Nigeria Institute of Professional Engineers and Scientists, NIPES.